

LaundroMaster — Full Technical Specification (Spec.md)

Project Name

LaundroMaster

Project Overview

LaundroMaster is a scalable multi-platform laundromat marketplace ecosystem designed to connect customers with nearby laundromats in real time, similar to Uber-style service discovery and booking.

The platform consists of:

1. LaundroMaster — Customer Mobile App
2. LaundroMaster Owner — Laundromat Owner Mobile App
3. LaundroMaster Admin Panel — Web-based administration and management portal

The system allows laundromat businesses to onboard onto the platform, upload verification documents, publish their services, receive bookings, manage orders, track earnings, and interact with customers.

Customers can discover laundromats nearby using an interactive map interface, place laundry bookings, make payments, track orders, review laundromats, and receive notifications.

Core Technologies

Layer	Technology
Mobile Framework	Expo React Native
Development Platform	Expo Snack
Routing	Expo Router
Backend APIs	Vercel Serverless Functions
Database	Supabase PostgreSQL
Authentication	Supabase Auth
File Storage	Supabase Storage

Layer	Technology
Payments	iKhokha
Maps	react-native-maps
Push Notifications	Expo Notifications
Realtime	Supabase Realtime
State Management	Zustand
Forms	React Hook Form
Validation	Zod
UI Components	React Native Paper
Hosting	Vercel
Admin Panel	HTML + TailwindCSS + Vanilla JavaScript

Platform Structure

```

LaundroMaster Ecosystem
|
├─ LaundroMaster (Customer App)
├─ LaundroMaster Owner (Owner App)
└─ LaundroMaster Admin Panel (Web)

```

System Objectives

Customer Objectives

Customers must be able to:

- Create accounts
- Locate nearby laundromats
- View laundromat details
- Compare ratings
- Book laundry services
- Schedule pickups
- Schedule deliveries
- Make secure payments

- Track order progress
- Receive notifications
- Chat with laundromats
- Leave reviews
- Save favorite laundromats

Laundromat Owner Objectives

Owners must be able to:

- Register laundromat businesses
- Upload verification documents
- Configure service pricing
- Manage bookings
- Track earnings
- Manage staff
- Respond to customers
- Accept or reject bookings
- View analytics
- Manage operating hours

Admin Objectives

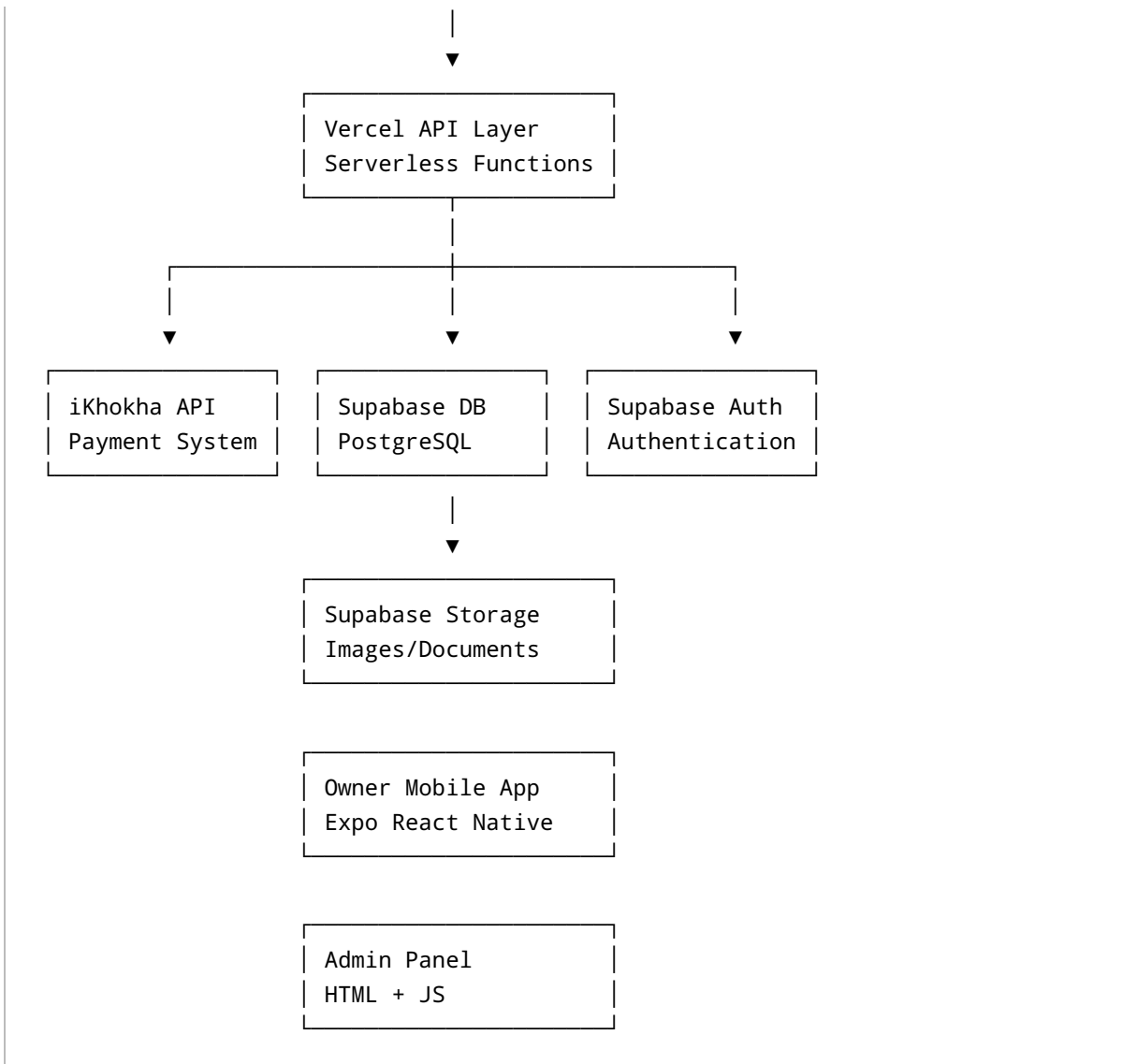
Admins must be able to:

- Vet laundromat registrations
- Approve or reject businesses
- Verify documents
- Suspend users
- Manage disputes
- Monitor payments
- View analytics
- Configure commissions
- Manage platform settings

High-Level Architecture

```
graph TD; A[Customer Mobile App  
Expo React Native];
```

Customer Mobile App
Expo React Native



Security Architecture

Security Objectives

Sensitive credentials must never be exposed directly inside Expo applications.

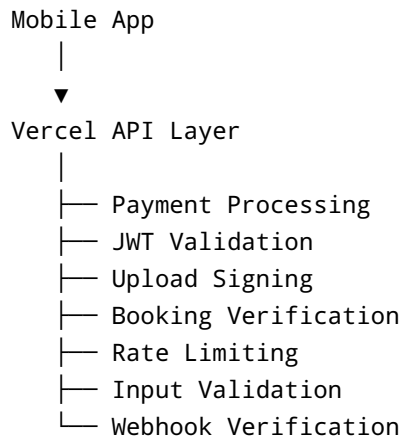
The following secrets must remain server-side only:

- Supabase service role key
- iKhokha secret keys
- Webhook secrets
- Admin credentials

- Internal API secrets
-

Vercel API Security Layer

All sensitive operations pass through Vercel serverless APIs.



Maximum API Structure

The system must not exceed 12 serverless API groups.

```
/api
├─ auth
├─ bookings
├─ payments
├─ laundromats
├─ uploads
├─ reviews
├─ notifications
├─ analytics
├─ payouts
├─ admin
├─ geo
└─ webhooks
```

API Responsibilities

auth

Handles:

- JWT validation
 - Session verification
 - Role authorization
-

bookings

Handles:

- Booking creation validation
 - Booking state updates
 - Booking security checks
-

payments

Handles:

- iKhokha payment initialization
 - Payment verification
 - Refund requests
-

laundromats

Handles:

- Sanitized laundromat retrieval
 - Nearby laundromat search
 - Filtering
 - Availability
-

uploads

Handles:

- Secure upload signatures
 - Image upload authorization
 - File validation
-

reviews

Handles:

- Review moderation
 - Rating aggregation
-

notifications

Handles:

- Push notification triggers
 - Notification delivery events
-

analytics

Handles:

- Revenue aggregation
 - Usage metrics
 - Reporting
-

payouts

Handles:

- Owner withdrawals
 - Earnings calculations
-

admin

Handles:

- Admin authentication
 - Vetting actions
 - Moderation
-

geo

Handles:

- Distance calculations
 - Radius filtering
 - Nearby searches
-

webhooks

Handles:

- iKhokha webhook verification
 - Payment confirmations
-

Database Specification

Database Provider

Supabase PostgreSQL

Main Database Tables

```
users
laundromats
services
bookings
booking_items
payments
reviews
```



```
favorites
notifications
documents
staff
transactions
withdrawals
admin_logs
chat_rooms
messages
service_categories
operating_hours
delivery_zones
```

Database Table Specifications

users

Stores all user accounts.

Field	Type
id	uuid
role	text
full_name	text
email	text
phone	text
avatar_url	text
created_at	timestamp

laundromats

Stores laundromat businesses.

Field	Type
id	uuid
owner_id	uuid

Field	Type
name	text
description	text
logo_url	text
address	text
latitude	decimal
longitude	decimal
rating	decimal
is_verified	boolean
is_open	boolean
created_at	timestamp

services

Stores laundromat services.

Field	Type
id	uuid
laundromat_id	uuid
name	text
description	text
base_price	decimal
turnaround_hours	integer

bookings

Stores booking records.

Field	Type
id	uuid

Field	Type
customer_id	uuid
laundromat_id	uuid
status	text
total_amount	decimal
pickup_required	boolean
delivery_required	boolean
scheduled_at	timestamp
created_at	timestamp

payments

Stores payment records.

Field	Type
id	uuid
booking_id	uuid
amount	decimal
payment_status	text
transaction_reference	text
created_at	timestamp

reviews

Stores customer reviews.

Field	Type
id	uuid
customer_id	uuid
laundromat_id	uuid

Field	Type
rating	integer
review_text	text
created_at	timestamp

Database Relationships

```

users
├─ laundromats
├─ bookings
├─ reviews
├─ favorites
└─ messages

laundromats
├─ services
├─ bookings
├─ reviews
├─ staff
└─ delivery_zones

bookings
├─ payments
├─ booking_items
└─ notifications

```

Authentication Specification

User Roles

Role	Access
customer	Customer app
owner	Owner app
admin	Admin panel

Authentication Methods

- Email and password
 - Phone OTP
 - Magic links (optional future support)
-

Authorization Rules

Customers

Can only access their own:

- bookings
 - reviews
 - chats
 - favorites
-

Owners

Can only access their own:

- laundromats
 - staff
 - bookings
 - analytics
 - payouts
-

Admins

Can access:

- platform-wide analytics
 - moderation systems
 - user management
 - business verification
-

Row Level Security (RLS)

All database tables must use Supabase Row Level Security.

Policies must:

- Restrict unauthorized reads
 - Restrict unauthorized writes
 - Enforce role-based access
 - Prevent cross-account access
-

Customer Mobile App Specification

App Name

LaundroMaster

Customer App Screens

```
Auth
├─ Splash
├─ Onboarding
├─ Login
├─ Register
├─ OTP Verification
└─ Forgot Password

Main
├─ Home
├─ Nearby Laundromats
├─ Map Discovery
├─ Search
├─ Laundromat Details
├─ Booking
├─ Checkout
├─ Orders
├─ Order Tracking
├─ Chat
├─ Notifications
└─ Favorites
```

- └─ Reviews
- └─ Wallet
- └─ Profile
- └─ Settings

Home Screen Features

The home screen must display:

- Search bar
- Nearby laundromats
- Featured laundromats
- Promotions
- Quick categories
- Recently used laundromats

Map Discovery Screen

The map screen must include:

- Interactive map
- User location
- Nearby laundromat markers
- Real-time filtering
- Radius filtering
- Open/closed indicators

Laundromat Cards

Each laundromat card must display:

- Avatar/logo
- Name
- Rating
- Total reviews
- Distance
- Open/closed status
- Pickup availability
- Delivery availability
- Service categories

- Estimated turnaround time
 - Price indicator
-

Booking Flow

```
graph TD; A[Select Laundromat] --> B[Select Services]; B --> C[Add Laundry Items]; C --> D[Choose Pickup/Delivery]; D --> E[Select Date & Time]; E --> F[Checkout]; F --> G[Payment]; G --> H[Booking Confirmation];
```

Checkout Features

The checkout screen must include:

- Order summary
 - Pricing breakdown
 - Taxes
 - Delivery fee
 - Promo codes
 - Payment method selection
-

Order Tracking

Customers must be able to track statuses such as:

- Pending
- Accepted
- Pickup Scheduled

- Washing
 - Drying
 - Ironing
 - Out for Delivery
 - Completed
-

Owner Mobile App Specification

App Name

LaundroMaster Owner

Owner App Screens

Auth

- └─ Login
- └─ Register
- └─ Verification
- └─ KYC Upload

Dashboard

- └─ Overview
 - └─ Orders
 - └─ Customers
 - └─ Services
 - └─ Pricing
 - └─ Earnings
 - └─ Analytics
 - └─ Staff
 - └─ Reviews
 - └─ Notifications
 - └─ Payouts
 - └─ Settings
-

Owner Dashboard Features

The dashboard must display:

- Daily revenue
 - Active orders
 - Pending orders
 - Weekly analytics
 - Customer satisfaction
 - Payout summaries
-

Verification Requirements

Owners must upload:

- Identity document
 - Business registration
 - Utility bill
 - Banking confirmation
-

Staff Management

Owners must be able to:

- Add staff
 - Remove staff
 - Assign permissions
 - Manage schedules
-

Admin Panel Specification

Admin Panel Technology

- HTML
 - TailwindCSS
 - Vanilla JavaScript
-

Admin Panel Pages

```
index.html
├─ Login
├─ Dashboard
├─ User Management
├─ Laundromat Vetting
├─ Verification Center
├─ Booking Monitoring
├─ Revenue Monitoring
├─ Disputes
├─ Reports
├─ Analytics
└─ System Settings
```

Admin Dashboard Features

Admins must be able to view:

- Total users
- Total laundromats
- Daily bookings
- Revenue metrics
- Pending verifications
- Fraud alerts

File Storage Specification

Storage Provider

Supabase Storage

Storage Structure

```
storage
├─ avatars
└─ laundromats
```

- └ documents
- └ bookings
- └ receipts
- └ reviews
- └ banners

File Upload Rules

Uploads must:

- Validate file type
- Validate file size
- Use signed uploads
- Prevent unauthorized access

Realtime Specification

Supabase Realtime Usage

Realtime functionality must support:

- Booking updates
- Chat messages
- Notifications
- Owner dashboard updates
- Order status changes

Payment Specification

Payment Gateway

iKhokha

Payment Flow

```
graph TD; A[Customer App] --> B[Vercel Payment API]; B --> C[iKhokha Payment Session]; C --> D[Customer Payment]; D --> E[Webhook Verification]; E --> F[Supabase Update]; F --> G[Booking Confirmation];
```

Payment Features

The payment system must support:

- Card payments
- EFT payments
- Refund requests
- Webhook verification
- Secure transaction references

Push Notification Specification

Notification Types

The system must support:

- Booking confirmations
- Order updates
- Promotions
- Chat messages
- Payment confirmations
- Verification updates

Theme Specification

Brand Colors

Color	HEX
Primary Red	#D90429
Black	#111111
Grey	#6C757D
White	#FFFFFF

UI Design Rules

The design system must use:

- Rounded corners
- Floating action buttons
- Soft shadows
- Minimalistic layouts
- Modern card interfaces
- Responsive spacing

Mobile Component Specification

Customer Components

```
components/  
├─ LaundromatCard  
├─ BookingCard  
├─ ServiceSelector  
├─ MapMarker  
├─ RatingStars  
├─ PriceBreakdown  
├─ BookingTimeline  
└─ OrderTracker
```

```
|— NotificationCard
|— ChatBubble
```

Owner Components

```
components/
|— RevenueCard
|— BookingManager
|— AnalyticsChart
|— PayoutCard
|— VerificationStatus
|— ServiceEditor
|— StaffManager
```

Customer App File Map

```
LaundroMaster/
|— app/
|   |— (auth)/
|   |— (tabs)/
|   |— booking/
|   |— laundromat/
|   |— checkout/
|   |— orders/
|   |— chat/
|   |— profile/
|   |— settings/
|   |
|   |— assets/
|   |   |— icons/
|   |   |— fonts/
|   |   |— images/
|   |   |— animations/
|   |
|   |— components/
|   |   |— cards/
|   |   |— buttons/
|   |   |— maps/
|   |   |— modals/
```

```
|   |─ forms/
|   |─ ratings/
|   └─ loaders/
|
|─ hooks/
|─ services/
|─ stores/
|─ utils/
|─ constants/
|─ theme/
|─ types/
|─ lib/
└─ config/
```

Owner App File Map

```
LaundroMasterOwner/
├─ app/
├─ components/
├─ analytics/
├─ services/
├─ hooks/
├─ stores/
├─ lib/
├─ theme/
├─ assets/
└─ config/
```

Admin Panel File Map

```
admin/
├─ index.html
├─ css/
├─ js/
├─ assets/
├─ pages/
└─ components/
```



```
├─ charts/  
└─ services/
```

Monorepo Structure

```
LaundroMaster-Ecosystem/  
├─ apps/  
│   ├── customer-app/  
│   ├── owner-app/  
│   └─ admin-panel/  
├─ shared/  
│   ├── ui/  
│   ├── constants/  
│   ├── theme/  
│   ├── types/  
│   └─ utils/  
├─ api/  
│   ├── auth/  
│   ├── bookings/  
│   ├── payments/  
│   ├── uploads/  
│   ├── webhooks/  
│   └─ analytics/  
├─ docs/  
├─ scripts/  
└─ package.json
```

Recommended Expo Packages

```
expo-router  
expo-location  
expo-notifications  
expo-image-picker  
react-native-maps  
react-native-map-clustering  
react-native-paper
```

```
react-hook-form
zod
zustand
react-native-svg
```

Performance Optimization

The system must implement:

- Lazy loading
 - Infinite scrolling
 - Skeleton loaders
 - Image compression
 - Offline caching
 - Map clustering
-

Scalability Goals

Initial scalability targets:

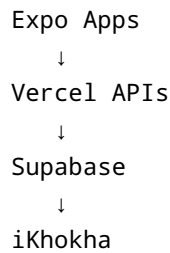
- 1,000 laundromats
 - 100,000 users
 - 10,000 concurrent bookings
-

Future Expansion Modules

Future modules may include:

- Driver delivery system
 - AI recommendations
 - Dynamic pricing
 - QR pickup verification
 - WhatsApp integration
 - Loyalty rewards
 - Subscription plans
 - Franchise support
 - Multi-country support
-

Deployment Architecture



```
graph TD; Expo[Expo Apps] --> Vercel[Vercel APIs]; Vercel --> Supabase[Supabase]; Supabase --> iKhokha[iKhokha];
```

Expo Apps
↓
Vercel APIs
↓
Supabase
↓
iKhokha

Critical Security Rules

The system must always:

- Use Row Level Security
- Validate JWT tokens
- Verify webhooks
- Sanitize inputs
- Use signed uploads
- Protect server-side secrets
- Enforce rate limiting

Conclusion

LaundroMaster is designed as a scalable, secure, marketplace-style laundromat ecosystem that combines real-time booking, interactive maps, secure payments, and multi-role management into a unified platform architecture.

The platform architecture prioritizes:

- scalability
- modularity
- security
- real-time updates
- maintainability
- performance
- modern mobile UX